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Nickel-catalyzed multicomponent 1,4-fluoroalkylcarbonylation of 1,3-enynes to access allenyl ketones

Bing-Hong Teng [a,b,1], Chang-Sheng Kuai [b,1], Yingying Zhao [a,**], Xiao-Feng Wu [b,c,d,*]

[a] School of Chemistry and Chemical Engineering, Liaoning Normal University, 850 Huanghe Road, Dalian, 116029 China
[b] Dalian National Laboratory for Clean Energy, Dalian Institute of Chemical Physics, Chinese Academy of Sciences, Dalian, 116023, Liaoning, China
[c] Leibniz-Institut für Katalyse e.V., Albert-Einstein-Straße 29a, 18059, Rostock, Germany
[d] College of Material Chemistry and Chemical Engineering, Key Laboratory of Organosilicon Chemistry and Material Technology, Ministry of Education, Hangzhou Normal University, Hangzhou, 311121, Zhejiang, China

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ABSTRACT

Allenyl ketones are widely distributed in various bioactive molecules which lead their selective synthesis continues a hot research topic. Currently, most of the developed protocols require multistep and stoichiometric reactions together with noble metal catalysts. Herein, we developed a practical methodology to access allenyl ketones straightforwardly via multi-component coupling of 1,3-enynes, fluoroalkyl halides, arylboronic acids, and CO with nickel as the catalyst. A series of fluoroalkyl unit containing allenyl ketones were produced in moderate to good yields. This new procedure features cheap catalytic system, relatively mild reaction conditions, broad substrate scope, excellent regioselectivity, and attractive synthetic transformations of the obtained products.

Allenyl ketones are potent structural units that are widely present in diverse natural products and biologically relevant molecules such as citroside A and ketosteroid isomerase inhibitor [1]. Furthermore, they are also versatile building blocks in synthetic chemistry to give further valued added chemicals [2]. Therefore, the selective synthesis of such unique frameworks has received considerable attention over the past decades. The conventional methods include ketonization of pre-formed allene precursors with aldehyde or carboxylic acid derivatives, the elimination of unsaturated ketones and Wittig reaction, etc. (Fig. 1b) [3–7]. Nevertheless, these approaches often require stoichiometric oxidants and/or metal reagents and multi-step operations. In addition, some of the needed precursors are air/moisture sensitive and complicated to prepare which further restrict the structure diversity of the products. In this context, the exploration of a general and practical alternative procedure to access allenyl ketones with readily available reagents is highly desirable.

Multi-component reactions, due to the high step- and atom-economy, enables a rapid entry to build up molecular complexity from three or even more types of starting materials in a one-pot manner. By taking the advantages of carbon monoxide (abundant, readily available from carbon-containing wasters, etc.), carbonylative multicomponent reaction offers ideal choice for the preparation of carbonylated compounds from readily available substates [8]. Without surprise, this type of strategy has also been explored in the synthesis of carbonyl-containing allenyl derivatives, such as palladium-catalyzed carbonylation of propargyl alcohols or esters to produce 2,3-allenoic acid derivatives (Fig. 1c) [9,10]. Alcohols and amines were applied as the nucleophilic reagents in these reactions, however, for the using of carbon-based nucleophile as the coupling partner to produce alkenyl ketone product has not been achieved until the work from Guo and co-workers who recently reported a nickel-catalyzed four-component carbonylation of CF_3-containing 1,3-enynes, arylboronic acids, and cyclobutanone oxime esters [11]. The reaction was performed under atmospheric pressure of CO and various tetra-substituted CF_3-allenyl ketones were produced in good yields and selectivity.

However, developing new carbonylation reaction to enrich the alkenyl ketone products based on inexpensive catalyst is still appealing. Given our continuous interests in the field of developing new catalytic

* Corresponding author. Dalian National Laboratory for Clean Energy, Dalian Institute of Chemical Physics, Chinese Academy of Sciences, Dalian 116023, Liaoning, China.
** Corresponding author.
E-mail address: xiao-feng.wu@catalysis.de (X.-F. Wu).
[1] These authors contributed equally to this work.

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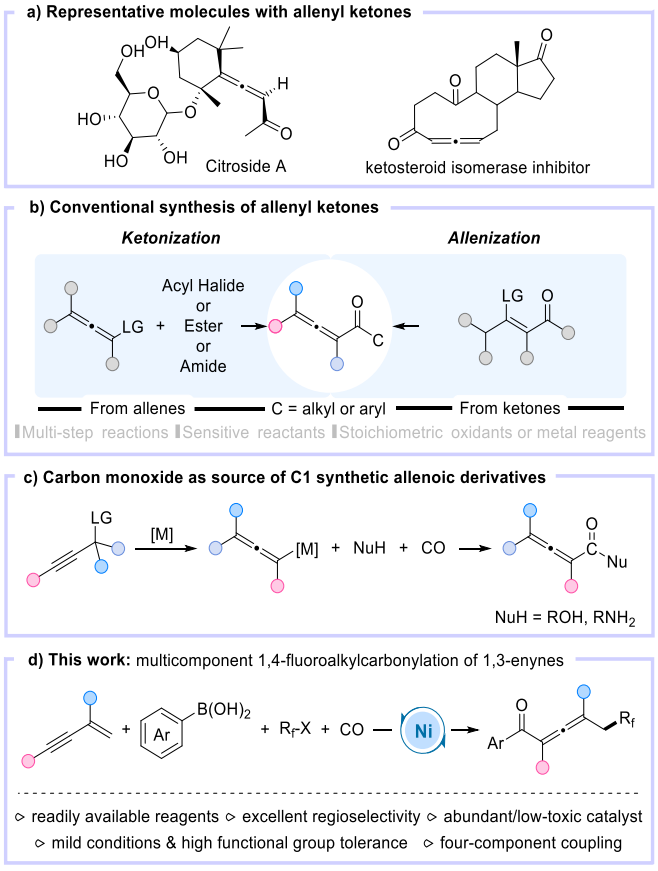
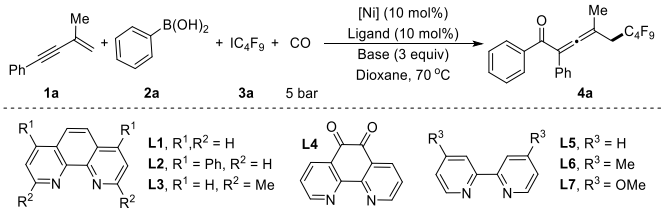


Fig. 1. Strategies for multicomponent 1,4-fluoroalkylcarbonylation.

carbonylation reactions, we herein present a nickel-catalyzed multicomponent coupling of 1,3-enynes, fluoroalkyl halides, arylboronic acids, and CO to afford structurally diverse allenyl ketones (Fig. 1d) [12]. An array of valuable fluoroalkyl containing allenyl ketones were formed in good yields in general. This new methodology also features wide functional group tolerance, excellent regioselectivity, readily available raw materials, inexpensive nickel catalyst and relatively mild reaction conditions.

To start the investigation, 1,3-enyne **1a** (0.2 mmol), phenylboronic acid **2a** (0.3 mmol) and perfluorobutyl iodide **3a** (0.3 mmol) were used as the model substrates. A range of nickel precatalysts were firstly screened using 1,10-phenanthroline (phen) as the ligand in 1,4-dioxane at 70 °C under 5 bar CO for 15 h. In the presence of $NiCl_2$·DME, the desired coupling could take place albeit with a slightly decreased yield (Table 1, entry 1). Other catalysts including NiI_2, $NiBr_2$ and $Ni(DPPE)Cl_2$ showed no catalytic activity (Table 1, entries 2–4). A better result was obtained when $Ni(OTf)_2$ (10 mol%) was used as the catalyst precursor (Table 1, entry 5). Compared to other inorganic bases such as K_2CO_3 and Na_3PO_4, Na_2CO_3 is a better choice here (Table 1, entry 7). The yield of allenyl ketone **4a** could be increased to 54% by extending the reaction time to 24 h (Table 1, entry 8). Considering the nature of the ligand is another important factor for tuning the reaction outcomes, several bis-nitrogen ligands were tested (**L1**-**L7**). The utilization of phenyl-substituted phen ligand **L2** resulted in an improvement in the yield of the desired product (Table 1, entry 9), while the bipyridine related ligands gave inferior results (Table 1, entries 10–14). To further increase the yield, the ratios of **1a**/**2a**/**3a** and Na_2CO_3 were screened (Table 1, entries 15–17). The results demonstrated that 86% yield could be achieved when the dosage of **2a** and Na_2CO_3 was increased to 2.0 and 4.0 equivalents, respectively (for more details, see Supporting Information). It should be noted that, only fluoroalkyl-derived allenyl ketone **4a** was

Table 1
Optimization of the Reaction Conditions[a].



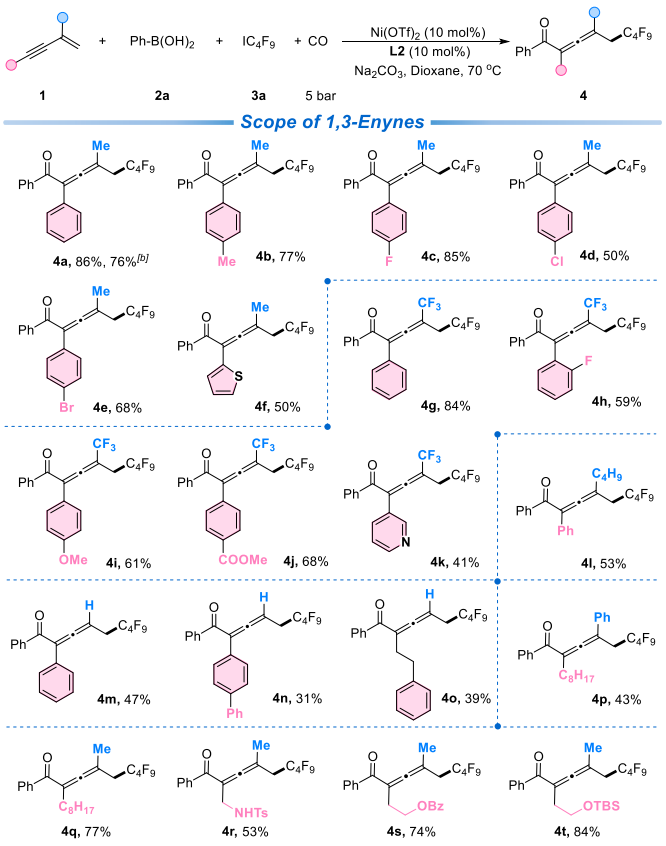
Entry	Catalyst	Ligand	Base	Yield (%)[b]
1	$NiCl_2$·DME	**L1**	K_2CO_3	30
2	NiI_2	**L1**	K_2CO_3	trace
3	$NiBr_2$	**L1**	K_2CO_3	trace
4	$Ni(DPPE)Cl_2$	**L1**	K_2CO_3	trace
5	$Ni(OTf)_2$	**L1**	K_2CO_3	32
6	$Ni(OTf)_2$	**L1**	Na_3PO_4	6
7	$Ni(OTf)_2$	**L1**	Na_2CO_3	33
8[c]	$Ni(OTf)_2$	**L1**	Na_2CO_3	54
9	$Ni(OTf)_2$	**L2**	Na_2CO_3	68
10	$Ni(OTf)_2$	**L3**	Na_2CO_3	7
11	$Ni(OTf)_2$	**L4**	Na_2CO_3	31
12	$Ni(OTf)_2$	**L5**	Na_2CO_3	12
13	$Ni(OTf)_2$	**L6**	Na_2CO_3	17
14	$Ni(OTf)_2$	**L7**	Na_2CO_3	27
15[d]	$Ni(OTf)_2$	**L2**	Na_2CO_3	79
16[d,e]	$Ni(OTf)_2$	**L2**	Na_2CO_3	78
17[df]	$Ni(OTf)_2$	**L2**	Na_2CO_3	87 (86)

[a] Reaction conditions: **1a** (0.2 mmol), **2a** (0.3 mmol), **3a** (0.3 mmol), [Ni] (10 mol%), ligand (10 mol%), Na_2CO_3 (3.0 equiv), 1,4-dioxane (1.0 mL), CO (5 bar), 70 °C, 15 h.
[b] Yield was determined by GC using *n*-hexadecane as the internal standard. The isolated yield is given in parentheses.
[c] 24 h.
[d] **2a** (0.4 mmol).
[e] **3a** (0.4 mmol).
[f] Na_2CO_3 (4.0 equiv).

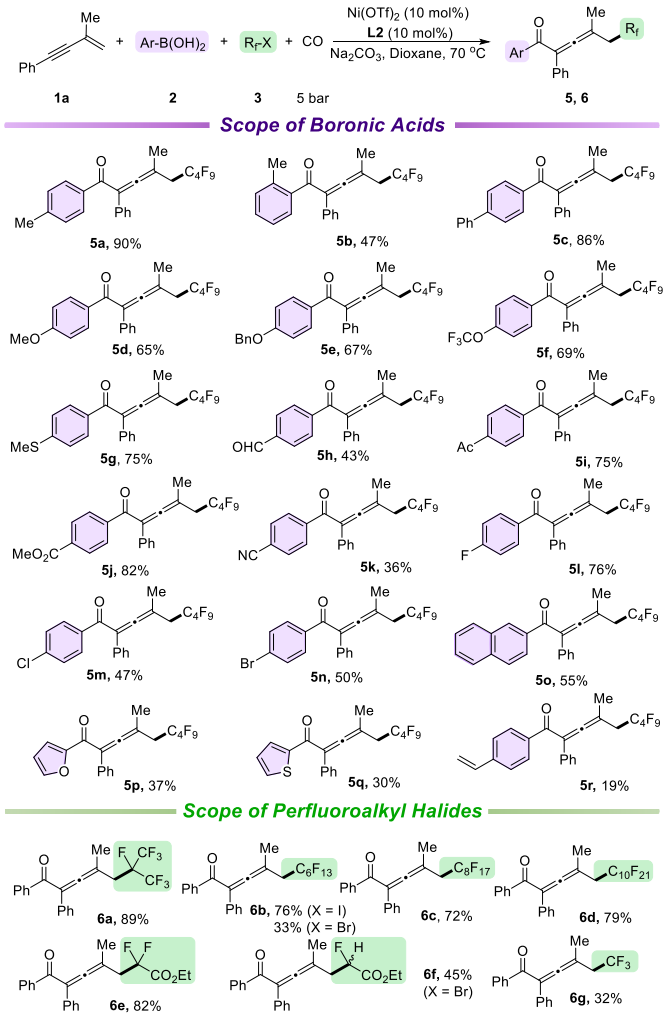
exclusively obtained in all these reactions and no other regioisomers were observed. No desired product could be detected under higher pressure of CO while 1 bar of CO lead to decreased yield.

Having established the optimal reaction conditions, we next turned our attention to explore the generality of this 1,4-fluoroalkylcarbonylation reaction. In the first stage, the scope with respect to various 1,3-enynes **1** with phenylboronic acid **2a** and perfluorobutyl iodide **3a** was examined (Scheme 1). An array of aromatic groups at the 4-position or different alkyl groups at the 2-position of 1,3-enynes all worked well to generate the corresponding products (**4a**-**4p**) in moderate to good yields. Moreover, various functional groups on 1,3-alkynes were also well tolerated in this catalytic system, including halides (**4c**-**4e**, **4h**), thiophene (**4f**), methoxyl (**4i**), ether (**4k**, **4j**) and pyridine (**4k**). To our delight, 1,3-enynes with a trifluoromethyl group at the 2-position did not occur *β*-F elimination (**4g**-**4k**). When there was no substituent at the 2-position, decreased yields were observed (**4m**-**4o**). This distinct reactivity may be due to the instability of the formed propargyl radical and the derived allenyl radical. Next, long chain octyl group and diverse heteroatom-substituted alkyl groups on the 4-position (**4p**-**4t**) of 1,3-enynes were applicable to this process as well. To demonstrate the versatility and practicality of this reaction, the model system was run on a 2.0 mmol scale under 5 bar of CO, achieving 76% yield of the desired product **4a**.

Subsequently, the scope of other coupling reagents, including arylboronic acids and perfluoroalkyl halides were examined (Scheme 2). Arylboronic acids bearing electron-donating groups (-Me, -Ph or –RO) at para- and ortho-positions could smoothly undergo the transformation to give the corresponding allenyl ketones (**5a**-**5f**) in 47–90% yields. The thioether (**5g**) unit with coordinating ability was compatible with the



Scheme 1. Substrate Scope of 1,3-Enynes[a]
[a]Reaction conditions: **1** (0.2 mmol), **2a** (0.4 mmol),**3a** (0.3 mmol), $Ni(OTf)_2$ (10 mol%), **L2** (10 mol%), Na_2CO_3 (4.0 equiv), 1,4-dioxane (1.0 mL), CO (5 bar), 70 °C, 24 h. Isolated yields.
[b]2 mmol scale.



Scheme 2. Substrate Scope of Arylboronic Acids and Perfluoroalkyl halides[a]
[a]Reaction conditions: **1a** (0.2 mmol), **2** (0.4 mmol), **3** (0.3 mmol), $Ni(OTf)_2$ (10 mol%), **L2** (10 mol%), Na_2CO_3 (4.0 equiv), 1,4-dioxane (1.0 mL), CO (5 bar), 70 °C, 24 h. Isolated yields.

reaction as well. Electron-withdrawing acyl and ester groups were well suitable with the process (**5i**, **5j**), while aldehyde and cyano groups were sensitive to nucleophilic reagents and thus decreased yields were obtained (**5h**, **5k**). Notably, halides such as F, Cl and Br were well tolerated and gave the desired products in moderate to good yields (**5l-5n**), which can offer the possibility for further derivatization via cross-coupling reactions. Moreover, naphthyl and heterocyclic boronic acids were also feasible in this reaction system (**5o-5q**). In the case of 4-vinylphenyl boronic acid, only 19% yield was attained, may be attributed that the double bond can have an effect on the radical addition step. Then, a range of perfluorinated and difluorinated halides, including sterically hindered and long chain iodides, participated in the reaction efficiently and the corresponding products (**6a-6e**) were afforded in moderate to high yields. Due to the increased bond energy, the reactions involving perfluorohexyl bromide and ethyl bromofluoroacetate (**6f**) resulted in decreased yields. When trifluoroiodomethane was used as a precursor of $\cdot CF_3$ radical, the expected product (**6g**) was also furnished in 32% yield.

To verify the synthetic potential of this nickel-catalyzed fluoroalkylcarbonylation reaction, additional efforts were directed towards Nazarov cyclization of the resulting allenyl ketones. A treatment of phenyl allenyl ketone **5a** with DMSO at 160 °C led to the formation of cyclized product **7a** in 63% yield with good stereoselectivity (Scheme 3). Furan- and methoxyl-substituted allenyl ketones were applicable to this intramolecular cyclization as well. However, it is worth noting that in the case of electron-withdrawing cyanide-derived allenyl ketone **5k**, domino cyclization and isomerization of carbon-carbon double bond executed, giving rise to indanone **7c** as the main product. To probe the reaction mechanism, radical inhibition experiments were carried out. The yield of **4a** gradually decreased with increasing the dosage of radical scavenger 2,2,6,6-tetramethylpiperidinooxy (TEMPO, 0–3 equiv) (Scheme 3, bottom). This result reveals that radical intermediates may be involved in this 1,4-fluoroalkylcarbonylation of enynes.

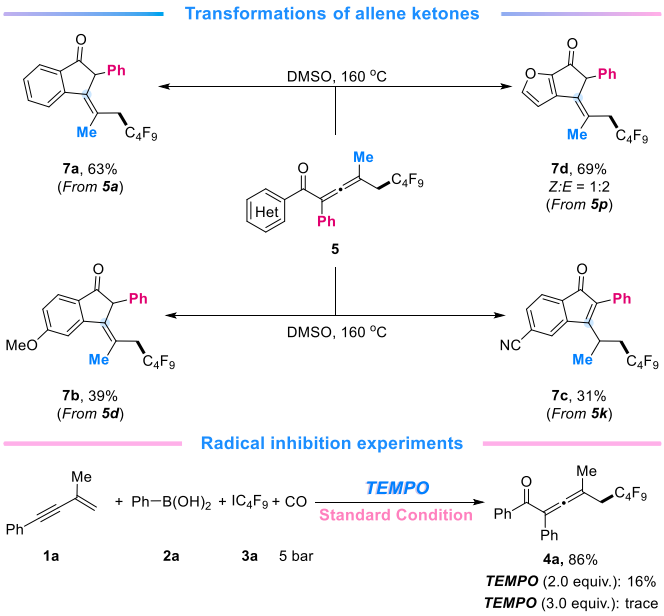
In conclusion, we have developed a practical multi-component coupling of 1,3-enynes, fluoroalkyl halides, arylboronic acids, and CO enabled by nickel catalysis.[15] This fluorocarbonylation process enables a rapid assembly of structurally diverse fluoroalkyl-derived allenyl ketones. This protocol features inexpensive catalytic system, mild reaction conditions, good functional group tolerance, and excellent regioselectivity. More importantly, the resulting products can also undergo intramolecular cyclization to furnish a variety of indanones.

CRediT authorship contribution statement

Bing-Hong Teng: Formal analysis, Data curation. **Chang-Sheng Kuai:** Formal analysis, Data curation. **Yingying Zhao:** Writing – original draft, Formal analysis. **Xiao-Feng Wu:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

We have no conflict of interest to declaration.



Scheme 3. Transformations of allenyl ketones and radical inhibition experiments.

Data availability

No data was used for the research described in the article.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.tet.2024.133965.

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